1. Software Overview

The software/interface is a detective game where you have to guess a number using hints from different files.
The number you enter can be seen on the screen.
If you enter the correct code, it shows that it is correct and then opens the box.

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2. System Logic

1. Use the website to start the game
2. Read the hints and guess the number
3. Enter the number using the rotary phone
4. The input is shown on the screen
5. If the number is correct, go to next step
6. Enter the Morse code using the IR sensor
7. If the Morse code is correct, it opens the hand
8. The box opens

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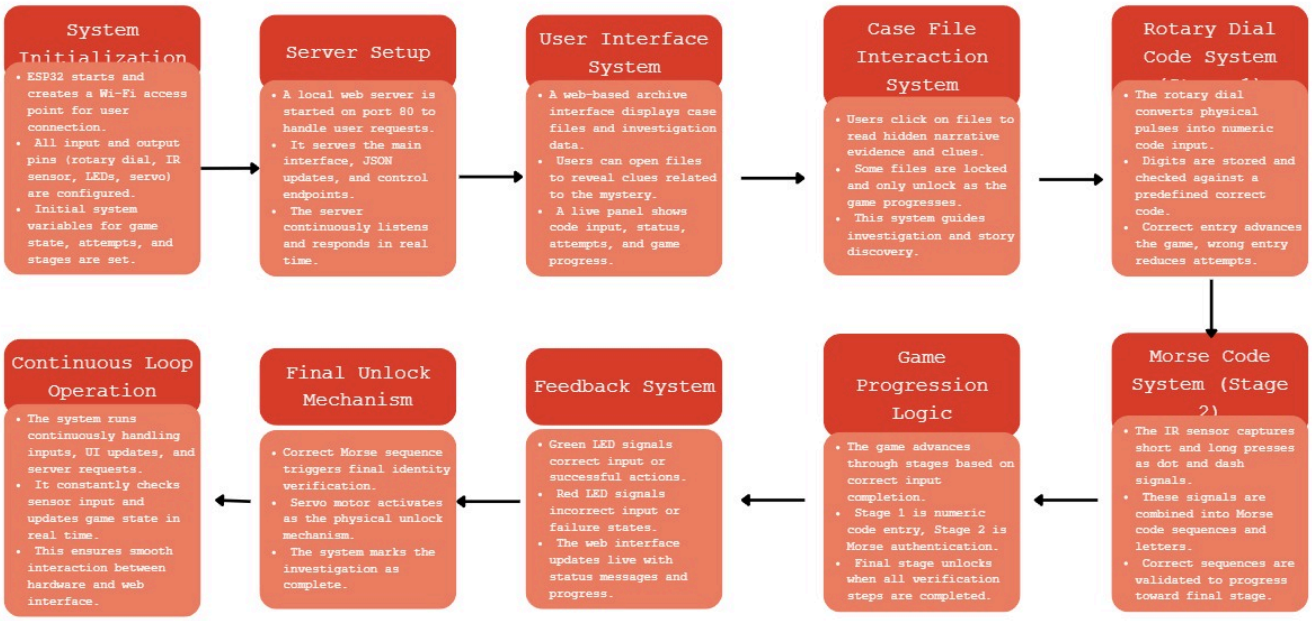
3. Flowchart

Start → Open website → See hints → Enter number → Check number →
If correct → Enter Morse code → Check Morse → Open box → End


System Initialization
• ESP32 starts and creates a Wi-Fi access point for user connection.
• All input and output pins (rotary dial, IR sensor, LEDs, servo) are configured.
• Initial system variables for game state, attempts, and stages are set.
Server Setup
• A local web server is started on port 80 to handle user requests.
• It serves the main interface, JSON updates, and control endpoints.
• The server continuously listens and responds in real time.
User Interface System
• A web-based archive interface displays case files and investigation data.
• Users can open files to reveal clues related to the mystery.
• A live panel shows code input, status, attempts, and game progress.
Case File Interaction System
• Users click on files to read hidden narrative evidence and clues.
• Some files are locked and only unlock as the game progresses.
• This system guides investigation and story discovery.
Rotary Dial Code System
• The rotary dial converts physical pulses into numeric code input.
• Digits are stored and checked against a predefined correct code.
• Correct entry advances the game, wrong entry reduces attempts.
Morse Code System (Stage 2)
• The IR sensor captures short and long presses as dot and dash signals.
• These signals are combined into Morse code sequences and letters.
• Correct sequences are validated to progress toward final stage.
Game Progression Logic
• The game advances through stages based on correct input completion.
• Stage 1 is numeric code entry, Stage 2 is Morse authentication.
• Final stage unlocks when all verification steps are completed.
Feedback System
• Green LED signals correct input or successful actions.
• Red LED signals incorrect input or failure states.
• The web interface updates live with status messages and progress.
Final Unlock Mechanism
• Correct Morse sequence triggers final identity verification.
• Servo motor activates as the physical unlock mechanism.
• The system marks the investigation as complete.
Continuous Loop Operation
• The system runs continuously handling inputs, UI updates, and server requests.
• It constantly checks sensor input and updates game state in real time.
• This ensures smooth interaction between hardware and web interface.

CODE FLOW